Raj Kumar Gupta

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Projects

Group Expense Management System | MERN Stack **Dec 2025 - Jan 2026**

- Developed a full-stack web application using **React, Node.js, and MongoDB** to facilitate seamless group expense tracking and splitting.
- Implemented secure **User Authentication** using **JWT** and HTTP-only cookies, ensuring robust data privacy and session management.
- Integrated **Razorpay API** to facilitate efficient in-app payments, enabling users to settle group balances directly via UPI, cards, and net banking.
- Implemented secure **CORS** policies and protected RESTful API endpoints to ensure safe data transmission between the client and server.

  GitHub Repository: **https://github.com/raj-kumar-96081/Expense-react-frontend**

SwiftRide: A Real-Time Ride-Sharing System | *MERN Stack, Socket.io, Google Maps API* **July 2025 - Aug 2025**

- Developed a full-stack platform with distinct workflows for **Users, Drivers, and Admins**, implementing secure **Role-Based Access Control (RBAC)** and a specialized driver onboarding pipeline.
- Designed a real-time matching engine using **Google Maps API** to perform distance-matrix calculations and automatically assign the nearest available driver to a user's pickup coordinates.
- Leveraged **Socket.io** for low-latency, bidirectional communication, ensuring instant ride request notifications and live status updates across all active interfaces.
- Built a centralized **Admin Dashboard** to manage user/driver databases and audit comprehensive ride histories, ensuring data integrity and system-wide operational oversight.

  GitHub Repository: **https://github.com/raj-kumar-96081/Ride-Sharing-Simulation-**

Trainings

Competitive Programming | AlgoTutor | *Certificate* **June 2025 - July 2025**

- Completed intensive 28-day training with hands-on problem-solving sessions focused on algorithms, data structures, and time-complexity optimization.
- Trained in advanced problem-solving techniques: **Recursion, DP, Greedy Algorithms, Trees, Graphs, STL.**
- Practiced daily coding assignments and competitive programming challenges on real contest patterns.
- Strengthened coding speed, Algorithmic optimization, and Structured thinking for Competitive Programming.

Certifications

Mastering Data Structure & Algorithms Using C/C++ **January, 2024**

Udemy — Certificate Link

Technical Skills

Languages: C++, Java, JavaScript, C, Python

Technologies/Frameworks: React.js, Node.js, Git, GitHub, Ubuntu, MySQL, MongoDB

Skills: Data Structures & Algorithms, Problem Solving, Responsive Web Design

Education

Lovely Professional University Punjab **2023 – Present**

Bachelor of Technology - Computer Science and Engineering- ***CGPA: 8.17*** *Phagwara, Punjab*

Guru Gobind Singh Public School **2020 – 2022**

12th with Science — Percentage: 89.60 *Ranchi, Jharkhand*

Guru Gobind Singh Public School **2019 – 2020**

10th with PCM — Percentage: 94.20 *Ranchi, Jharkhand*